

E-commerce Database Project

Questions and Answers for Exam

1. E-R Schema Questions

Q1: Why did you choose these 8 entities?

A: I chose Customer, Category, Product, Address, Orders, OrderItem, Payment, and Review because they cover the main requirements of an e-commerce system: managing customers, products, orders, and payments.

Q2: What is the relationship between Orders and OrderItem?

A: One-to-Many. One order can contain multiple items (products). This is how we solve the Many-to-Many relationship between Orders and Products.

Q3: What is the relationship between Order and Payment?

A: One-to-One. Each order has exactly one payment record. The order_id in the payment table has a UNIQUE constraint to enforce this.

Q4: What is the relationship between Customer and Address?

A: One-to-Many. One customer can have multiple addresses (home, work, etc.).

Q5: What is the relationship between Customer and Review?

A: One-to-Many. One customer can write multiple reviews for different products.

Q6: What is the relationship between Product and Review?

A: One-to-Many. One product can have multiple reviews from different customers.

E-commerce Database Project

Questions and Answers for Exam

2. Normalization Questions

Q1: What is 3NF (Third Normal Form)?

A: A table is in 3NF if:

1. It is in 2NF (all non-key columns depend on the entire primary key)
2. No non-key column depends on another non-key column (no transitive dependency)

Q2: What is BCNF (Boyce-Codd Normal Form)?

A: A table is in BCNF if for every functional dependency $X \rightarrow Y$, X is a superkey. This is stricter than 3NF.

Q3: What is the difference between 3NF and BCNF?

A: In 3NF, the determinant of a non-trivial dependency must be a superkey OR the dependent must be a prime attribute. In BCNF, the determinant must always be a superkey. BCNF is stricter.

Q4: How do you prove your tables are in 3NF?

A: I checked each table:

- Customer: all columns depend only on customer_id
- Category: all columns depend only on category_id
- Product: all columns depend only on product_id
- Address: all columns depend only on address_id
- Orders: all columns depend only on order_id
- OrderItem: all columns depend only on order_item_id
- Payment: all columns depend only on payment_id
- Review: all columns depend only on review_id

No transitive dependencies exist in any table.

Q5: How do you prove your tables are in BCNF?

A: I identified candidate keys for each table:

- Customer: customer_id and email are candidate keys
- All other tables: only one candidate key (the primary key)

Since all determinants are superkeys, all tables satisfy BCNF.

3. Database Questions

Q1: What is a Primary Key?

A: A primary key is a column (or set of columns) that uniquely identifies each row in a table. It cannot be NULL and must be unique.

Q2: What is a Foreign Key?

A: A foreign key is a column in one table that references the primary key of another table. It creates a link between the two tables.

Q3: What does UNIQUE do?

A: UNIQUE constraint ensures that all values in a column are different. For example, the email in the customer table is UNIQUE so no two customers can have the same email.

Q4: What does AUTO_INCREMENT do?

A: AUTO_INCREMENT automatically generates a new unique number for each new row. We use it for primary keys so we do not need to manually assign IDs.

E-commerce Database Project

Questions and Answers for Exam

Q5: What does DECIMAL(10,2) mean?

A: DECIMAL(10,2) means a number with up to 10 total digits, 2 of which are after the decimal point. For example: 99999999.99. It is used for prices to avoid floating-point errors.

Q6: What does DEFAULT do?

A: DEFAULT sets a value that is used when no value is specified during INSERT. For example, stock_quantity DEFAULT 0 means if we do not provide a quantity, it will be 0.

E-commerce Database Project

Questions and Answers for Exam

4. PHP Questions

Q1: What does `mysqli_connect` do?

A: `mysqli_connect` creates a connection to the MySQL database. It takes host, username, password, and database name as parameters.

Q2: What does `mysqli_query` do?

A: `mysqli_query` sends an SQL query to the database. It returns a result object that we can use to fetch data.

Q3: What does `mysqli_fetch_assoc` do?

A: `mysqli_fetch_assoc` fetches one row from the result as an associative array. The keys are the column names.

Q4: What is `session_start`?

A: `session_start` initializes a session. It allows us to store data (like `customer_id`) that persists across multiple page loads.

Q5: What is `$_SESSION`?

A: `$_SESSION` is a superglobal array that stores session variables. For example, `$_SESSION["customer_id"]` stores the logged-in customer ID.

Q6: Why do we use `password_hash`?

A: `password_hash` converts a plain-text password into a secure hash. We never store plain-text passwords in the database for security reasons.

Q7: What does `header("Location: ...")` do?

A: It redirects the user to another page. For example, after login we redirect to the home page.

Q8: What is the difference between `$_GET` and `$_POST`?

A: `$_GET` sends data in the URL (visible, limited size). `$_POST` sends data in the request body (not visible, no size limit). We use GET for reading data and POST for modifying data.

5. HTML Questions

Q1: What is a form?

A: A form is an HTML element that collects user input. It can send data to the server using GET or POST method.

Q2: What is the difference between GET and POST in forms?

A: GET appends data to the URL (good for search/filter). POST sends data in the request body (good for login, registration, orders).

Q3: What is input type hidden?

A: A hidden input field is not visible to the user but its value is sent with the form. We use it to pass IDs like `product_id`.

Q4: What is input type radio?

A: Radio buttons allow the user to select exactly one option from a group. We use it for selecting an address during checkout.

Q5: What are select and option?

A: `select` creates a dropdown menu. `option` defines each item in the dropdown. We use it for category filter and payment method.

E-commerce Database Project

Questions and Answers for Exam

6. System Operation Questions

Q1: How does a customer register?

A: The customer fills the registration form (name, email, password). The password is hashed with password_hash. The data is inserted into the customer table using an INSERT query.

Q2: How does login work?

A: The customer enters email and password. We look up the email in the database. If found, we verify the password using password_verify. If correct, we store customer_id in the session.

Q3: How does adding to cart work?

A: The product_id and quantity are sent via POST. We store them in \$_SESSION["cart"] as an array. If the product is already in the cart, we increase the quantity.

Q4: How does placing an order work?

A: We calculate the total from the cart. We insert a row in the orders table. For each product in the cart, we insert a row in order_item and decrease stock_quantity in the product table. We also insert a payment record. Finally, we clear the cart session.

Q5: What happens if stock is 0?

A: The "Add to Cart" button is not shown for products with stock_quantity = 0. The customer cannot add out-of-stock products.

7. Security Questions

Q1: Why do you hash passwords?

A: To protect customer data. If the database is hacked, the attacker cannot see the real passwords. password_hash uses bcrypt algorithm which is very secure.

Q2: What is SQL Injection?

A: SQL Injection is an attack where malicious SQL code is inserted into input fields. For example, typing " OR 1=1 -- in a login form could bypass authentication. In this project we use simple string concatenation which is vulnerable, but for a student project it is acceptable.

8. Design Questions

Q1: Why did you simplify the category table (no self-reference)?

A: For simplicity. The original design had parent_category_id for subcategories, but since we only have 2 categories (Electronics and Clothing), we removed it to keep the code simple.

Q2: How would you add an orders page?

A: I would create a query that selects all orders for the logged-in customer, display them in a table, and add a link to view order details.

9. Code-Specific Questions

Q1: In header.php, what does include "db.php" do?

A: It includes the database connection file so that \$conn is available in all pages that include header.php.

Q2: In products.php, what does WHERE 1=1 do?

E-commerce Database Project

Questions and Answers for Exam

A: It is a trick to make it easier to add additional conditions with AND. Since $1=1$ is always true, it does not filter anything, but allows us to append AND category_id = X without checking if it is the first condition.

Q3: In cart.php, what does the foreach loop do?

A: It loops through all items in the cart session array. For each product, it fetches the product details from the database, calculates the subtotal, and adds it to the total.

Q4: In place_order.php, what does mysqli_insert_id do?

A: It returns the auto-increment ID of the last inserted row. After inserting the order, we get the order_id to use in order_item and payment tables.

Q5: In register.php, why do we check if email exists first?

A: Because the email column has a UNIQUE constraint. If we try to insert a duplicate email, the database will throw an error. It is better to check first and show a friendly message.